

But the electron only travels half a circle, so time spent = $1/2 = 6.0 \times 10^{-12}$ s.

24 C

Wire XY is perpendicular to the magnetic field of the solenoid, so magnetic force acting

$$\text{on XY} = B_{\text{solenoid}} I_{\text{XY}} L_{\text{XY}}$$

$$= (\mu_0 n_{\text{solenoid}} I_{\text{solenoid}}) I_{\text{XY}} L_{\text{XY}}$$

$$= (4\pi \times 10^{-7} \times \frac{20000}{1.2} \times 3.0) \times 1.5 \times 2.0 \times 10^{-2}$$

$$= 1.9 \times 10^{-3} \text{ N}$$

25 B